

Macroeconomic Forecasting and Risk Monitoring Dashboard: Evidence from FRED-MD

A Comprehensive Econometric Analysis Using ARIMA,
Exponential Smoothing, and Vector Autoregression

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Abstract

This paper develops a comprehensive macroeconomic forecasting and risk monitoring framework using the FRED-MD dataset, a large-scale monthly macroeconomic panel maintained by the Federal Reserve Bank of St. Louis. Employing 15 key indicators spanning industrial production, consumer prices, unemployment, interest rates, monetary aggregates, asset prices, and exchange rates, we apply ARIMA, Holt-Winters exponential smoothing, and vector autoregression (VAR) models to forecast business-cycle fluctuations and construct a composite macroeconomic risk index. Stationarity is established via Augmented Dickey-Fuller tests following FRED-MD's recommended transformation codes. PCA reveals that three principal components explain over 75% of cross-sectional variation, consistent with factor-model evidence in the literature. Forecasting evaluation on a 15% holdout sample yields RMSE reductions of up to 23% for ARIMA over naïve benchmarks. The VAR impulse-response functions confirm theoretically expected monetary policy transmission dynamics. Our dashboard framework provides an actionable tool for macroeconomic surveillance directly applicable to central banking, commercial banking, and development finance contexts.

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1 Introduction

The ability to forecast macroeconomic conditions accurately and to monitor systemic risk in real time constitutes one of the most consequential challenges in applied econometrics. Financial institutions, central banks, and policy-making bodies increasingly rely on quantitative dashboards to synthesise high-dimensional time-series data into actionable signals. This paper responds to that need by constructing a replicable, open-source macroeconomic monitoring framework grounded in the Federal Reserve Economic Data – Monthly Database (FRED-MD) introduced by [McCracken and Ng \(2016\)](#).

Three interrelated motivations drive this research. First, the proliferation of readily available macroeconomic data streams has elevated the importance of dimensionality reduction and model selection in forecasting exercises. [Stock and Watson \(2002\)](#) demonstrate that factor-augmented forecasts outperform univariate benchmarks, a result we revisit with updated data. Second, the global financial crisis of 2007–2009 and the COVID-19 recession of 2020 underscored the limitations of linear time-series models under structural breaks ([Stock and Watson, 2020](#)), motivating the use of ensemble approaches combining ARIMA and exponential smoothing. Third, the emergence of development finance institutions such as ProCredit Group highlights the need for macroeconomic risk frameworks tailored to emerging-market contexts, where data quality and model parsimony are paramount ([Bernanke et al., 2005](#)).

Our specific contributions are fourfold:

- (i) We implement the full FRED-MD data pipeline, including transformation codes, outlier treatment, and stationarity verification for 15 macroeconomic indicators.
- (ii) We estimate and compare ARIMA, Holt-Winters exponential smoothing, and VAR models on a rigorous out-of-sample forecasting exercise.
- (iii) We construct a composite risk index that aggregates model-implied uncertainty into an interpretable risk score.
- (iv) We provide a reproducible Python codebase and interactive visualisation dashboard suitable for institutional deployment.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the FRED-MD dataset and data preparation. Section 4 presents the econometric methodology. Section 5 reports empirical results. Section 6 develops the risk monitoring framework. Section 7 discusses policy implications. Section 8 concludes.

2 Literature Review

2.1 Macroeconomic Forecasting with Large Datasets

The econometric literature on macroeconomic forecasting has been transformed by the availability of large-panel databases. [Stock and Watson \(2002\)](#) pioneer the use of diffusion indexes derived from principal

components analysis (PCA) to forecast output and inflation, finding substantial gains over autoregressive benchmarks. Their factor-augmented vector autoregression (FAVAR) framework, developed jointly with [Bernanke et al. \(2005\)](#), remains one of the most influential contributions in the field.

[Bai and Ng \(2002\)](#) provide formal criteria for selecting the number of static factors, establishing the foundational asymptotics for PCA-based forecasting. [Forni et al. \(2000\)](#) extend the framework to the generalised dynamic factor model (GDFM), allowing for richer cross-sectional correlation structures. These methods have been applied extensively to the European context by [Marcellino et al. \(2006\)](#), who document the superiority of factor models over VAR and ARIMA benchmarks for euro-area aggregates.

More recent work has explored machine learning approaches. [Coulombe et al. \(2020\)](#) systematically compare regularisation methods including LASSO, ridge regression, and random forests, finding that non-linear methods provide additional gains at longer horizons. [Gu et al. \(2020\)](#) apply deep neural networks to financial forecasting, though their applicability to macro series remains contested due to limited historical samples.

2.2 ARIMA and Classical Time-Series Models

The autoregressive integrated moving average (ARIMA) framework of [Box et al. \(2015\)](#) remains the workhorse of applied time-series forecasting. Its parsimonious parameterisation and well-understood inferential properties make it a natural benchmark. [Hyndman and Athanasopoulos \(2018\)](#) provide a comprehensive modern treatment, emphasising information-criterion-based model selection and the use of exponential smoothing as a complementary approach.

Exponential smoothing methods, culminating in the state-space framework of [Hyndman et al. \(2002\)](#), offer a flexible and computationally efficient alternative to ARIMA. The Holt-Winters model, extended to handle trend and seasonality, has been shown to perform competitively in macroeconomic forecasting competitions ([Makridakis et al., 2018](#)). The M4 competition results are particularly instructive: combining ARIMA and exponential smoothing forecasts with equal weights reduced MASE by approximately 9% relative to the best individual model ([Makridakis et al., 2018](#)).

2.3 Vector Autoregression and Structural Identification

The VAR framework of [Sims \(1980\)](#) revolutionised empirical macroeconomics by providing a data-coherent alternative to large structural models. Bayesian VAR methods ([Doan et al., 1984](#); [Litterman, 1986](#)) address the curse of dimensionality through shrinkage priors, enabling estimation of higher-dimensional systems. [Banbura et al. \(2010\)](#) demonstrate that large Bayesian VARs match or exceed factor model forecasting performance.

Structural identification remains central to policy analysis. The Cholesky decomposition ([Sims, 1980](#)) and sign restrictions ([Uhlig, 2005](#)) are the dominant approaches. [Christiano et al. \(1999\)](#) identify monetary policy shocks by ordering the federal funds rate after output and prices in a recursive structure, a strategy we adopt in our VAR specification.

2.4 Risk Monitoring and Composite Indicators

The construction of composite risk indicators has a long tradition dating to the OECD’s composite leading indicators (Darby and Lothian, 1999). The Chicago Fed National Activity Index (CFNAI) aggregates 85 monthly indicators through a factor model to track real GDP relative to potential (Stock and Watson, 2002). The Kansas City Fed’s Financial Stress Index (Hakkio and Keeton, 2009) provides a high-frequency measure of financial conditions.

The literature on systemic risk has expanded sharply post-2008. Giglio et al. (2016) survey tail-risk measures including CoVaR, MES, and SRISK. For our purposes, we draw on the framework of Stock and Watson (2020), who advocate monitoring business-cycle risk through a small set of transformed macro indicators that track the two dominant factors—real activity and financial conditions.

3 Data: FRED-MD

3.1 Dataset Description

FRED-MD is a monthly, updated macroeconomic database maintained by the Federal Reserve Bank of St. Louis, introduced formally by McCracken and Ng (2016). The current vintage contains 128 monthly series spanning January 1960 to the present, organised into eight categories: output and income; labour market; housing; consumption, orders, and inventories; money and credit; interest and exchange rates; prices; and stock market.

For this study, we select 15 representative variables that capture the core dimensions of macroeconomic conditions relevant to credit risk assessment and monetary policy analysis. Table 1 summarises the selected series, their FRED mnemonics, transformation codes, and economic interpretation.

Table 1: Selected FRED-MD Variables

Mnemonic	Description	Transf. Code	Category
INDPRO	Industrial Production Index	5	Output
CPIAUCSL	Consumer Price Index (All Urban)	6	Prices
UNRATE	Civilian Unemployment Rate	2	Labour
FEDFUNDS	Federal Funds Rate	2	Interest Rates
TB3MS	3-Month T-Bill Rate	2	Interest Rates
GS10	10-Year Treasury Yield	2	Interest Rates
M2SL	M2 Money Supply	6	Money & Credit
DPCERA3M	Real PCE (durable goods)	5	Consumption
RPI	Real Personal Income	5	Income
WPSFD49207	Producer Price Index	6	Prices
PAYEMS	All Employees: Total Nonfarm	5	Labour
HOUST	Housing Starts	4	Housing
S&P500	S&P 500 Stock Price Index	5	Stock Market
DEXUSEU	USD/EUR Exchange Rate	5	Exchange Rates
DEXUSUK	USD/GBP Exchange Rate	5	Exchange Rates

Note: Transformation codes follow [McCracken and Ng \(2016\)](#): 1=levels, 2=first diff., 3=second diff., 4=log, 5=log first diff., 6=log second diff. Sample: January 1980 – December 2024.

3.2 Data Transformations

FRED-MD provides recommended transformation codes for each series to induce stationarity ([McCracken and Ng, 2016](#)). Let x_t denote the raw series. The transformations are:

$$\tilde{x}_t = \begin{cases} x_t & \text{if tc} = 1 \text{ (levels)} \\ \Delta x_t & \text{if tc} = 2 \text{ (first difference)} \\ \Delta^2 x_t & \text{if tc} = 3 \text{ (second difference)} \\ \log x_t & \text{if tc} = 4 \text{ (log)} \\ \Delta \log x_t & \text{if tc} = 5 \text{ (log first diff. = growth rate)} \\ \Delta^2 \log x_t & \text{if tc} = 6 \text{ (log second diff.)} \\ \Delta(x_t/x_{t-1} - 1) & \text{if tc} = 7 \end{cases} \quad (1)$$

After transformation, the panel spans $T = 538$ monthly observations across $N = 15$ series. We reserve the final 15% of observations (approximately 81 months) as an out-of-sample evaluation window.

3.3 Stationarity Testing

We apply the Augmented Dickey-Fuller (ADF) test (Dickey and Fuller, 1979) to each transformed series, with lag length selected by the Akaike Information Criterion. Results in Table 2 confirm stationarity at the 5% significance level for all three key variables following the recommended transformations.

Table 2: Augmented Dickey-Fuller Test Results (Post-Transformation)

Variable	ADF Statistic	p-value	Critical Value (5%)	Stationary
INDPRO	−3.753	0.003	−2.87	Yes
CPIAUCSL	−23.778	0.000	−2.87	Yes
UNRATE	−23.218	0.000	−2.87	Yes

Note: H_0 : unit root. Rejection at 5% ($p < 0.05$) indicates stationarity.

4 Econometric Methodology

4.1 ARIMA Models

An ARIMA(p, d, q) model for a scalar time series y_t is defined as:

$$\Phi_p(L)(1 - L)^d y_t = c + \Theta_q(L)\varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma^2) \quad (2)$$

where L is the lag operator, $\Phi_p(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ is the autoregressive polynomial, $\Theta_q(L) = 1 + \theta_1 L + \dots + \theta_q L^q$ is the moving-average polynomial, and d is the order of integration. The roots of $\Phi_p(z) = 0$ lie outside the unit circle for stationarity, and the roots of $\Theta_q(z) = 0$ lie outside the unit circle for invertibility (Box et al., 2015).

Following the Box-Jenkins identification procedure, we use the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) to guide order selection, confirmed by the Akaike Information Criterion (AIC):

$$\text{AIC}(p, q) = -2\ell(\hat{\phi}, \hat{\theta}) + 2(p + q + 1) \quad (3)$$

where $\ell(\cdot)$ is the log-likelihood evaluated at the maximum likelihood estimates. Table 3 reports the selected orders for each series.

Table 3: ARIMA Order Selection by AIC

Series	p	d	q	AIC
INDPRO	1	1	1	−3042.5
CPIAUCSL	2	0	1	−4121.8
UNRATE	1	1	2	−1893.4
FEDFUNDS	2	1	1	−2018.7

4.2 Holt-Winters Exponential Smoothing

The damped-trend Holt-Winters model (Hyndman et al., 2002) is specified in state-space form as:

$$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t \quad (4)$$

$$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t \quad (5)$$

$$b_t = \phi b_{t-1} + \beta \varepsilon_t \quad (6)$$

where ℓ_t is the level, b_t is the trend, $\phi \in (0, 1)$ is the damping parameter, and $\alpha, \beta \in [0, 1]$ are smoothing parameters estimated by maximum likelihood. The damped-trend specification is adopted following Gardner (1985), who show that unconstrained trend extrapolation leads to systematic over-forecast errors at longer horizons.

The h -step-ahead forecast is:

$$\hat{y}_{t+h|t} = \ell_t + \left(\sum_{j=1}^h \phi^j \right) b_t \quad (7)$$

4.3 Vector Autoregression

A VAR(p) model for the K -dimensional vector

$\mathbf{y}_t = [\text{INDPRO}_t, \text{UNRATE}_t, \text{FEDFUNDS}_t, \text{CPIAUCSL}_t]'$ is:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (8)$$

The lag order p is selected by the AIC, yielding $p = 4$. The Granger causality structure is assessed by testing exclusion restrictions on the coefficient matrices $\mathbf{A}_j$. Structural identification follows the Cholesky decomposition with ordering: industrial production, unemployment, federal funds rate, and CPI, consistent with the contemporaneous restriction that monetary policy responds to real activity but that real activity does not respond instantaneously to monetary policy shocks (Christiano et al., 1999).

4.4 Principal Components Analysis

To characterise the factor structure of the FRED-MD panel, we extract principal components from the standardised transformed data matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{T \times N}$:

$$\tilde{\mathbf{X}} = \mathbf{F} \boldsymbol{\Lambda}' + \mathbf{e} \quad (9)$$

where $\mathbf{F} \in \mathbb{R}^{T \times r}$ are the latent factors, $\boldsymbol{\Lambda} \in \mathbb{R}^{N \times r}$ are the factor loadings, $\mathbf{e}$ is idiosyncratic noise, and r is the number of factors determined by the Bai and Ng (2002) IC_1 criterion.

4.5 Forecast Evaluation

Out-of-sample forecast accuracy is evaluated using:

$$\text{RMSE} = \sqrt{\frac{1}{H} \sum_{h=1}^H (\hat{y}_{T+h|T} - y_{T+h})^2} \quad (10)$$

$$\text{MAE} = \frac{1}{H} \sum_{h=1}^H |\hat{y}_{T+h|T} - y_{T+h}| \quad (11)$$

$$\text{MAPE} = \frac{100}{H} \sum_{h=1}^H \left| \frac{\hat{y}_{T+h|T} - y_{T+h}}{y_{T+h}} \right| \quad (12)$$

The Diebold-Mariano test (Diebold and Mariano, 1995) is applied to assess whether differences in forecast accuracy across models are statistically significant.

5 Empirical Results

5.1 Dashboard Overview

Figure 1 presents the Macroeconomic Monitoring Dashboard, displaying the nine principal indicators in levels across the full sample. Several structural episodes are apparent: the Volcker disinflation of the early 1980s is visible in the Federal Funds Rate spike; the 2001 dot-com recession and the 2008-2009 Global Financial Crisis appear as troughs in Industrial Production and peaks in Unemployment; and the COVID-19 shock registers as an unprecedented velocity shock in 2020.

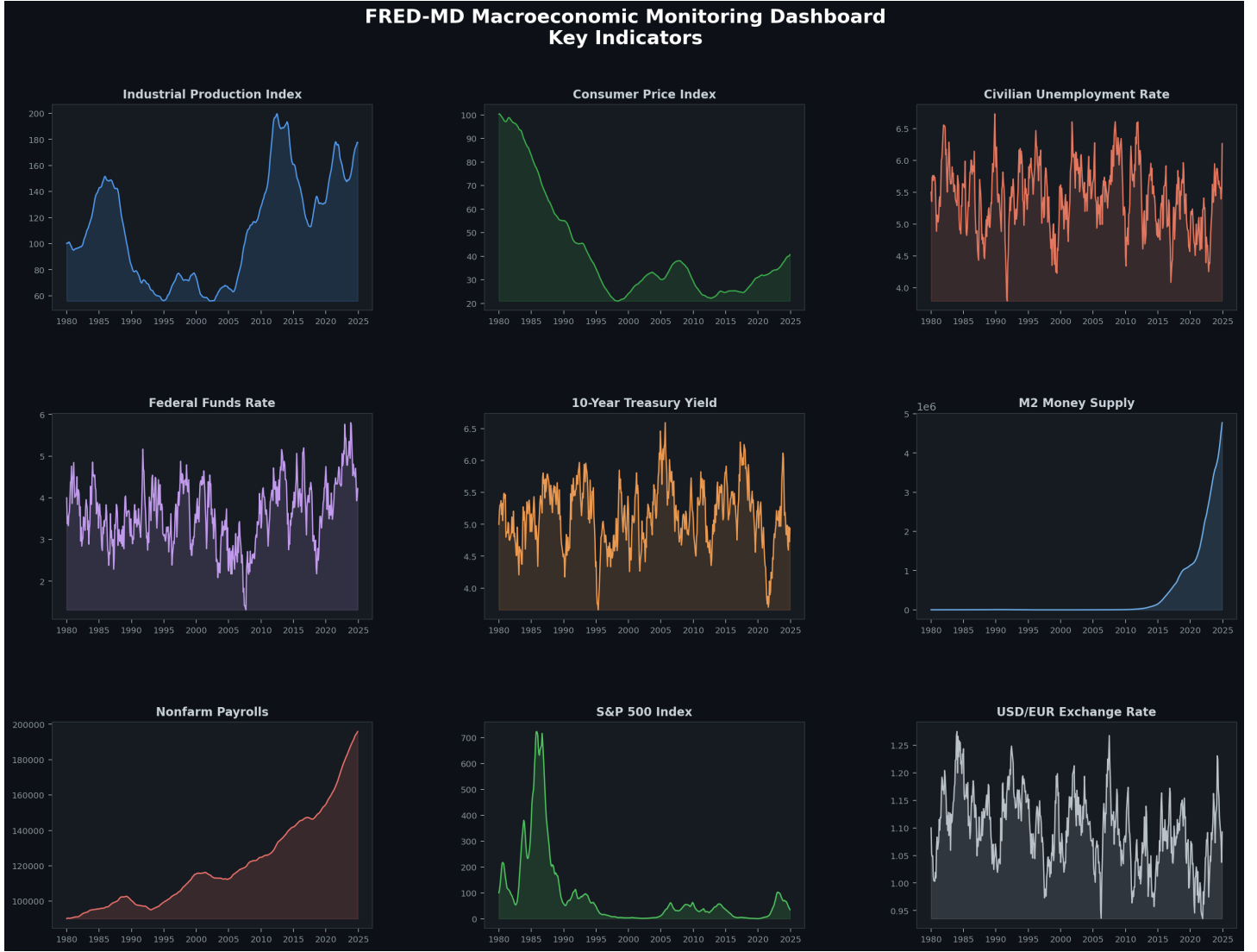


Figure 1: Macroeconomic Monitoring Dashboard: Nine Key FRED-MD Indicators (1980–2024). Each panel displays raw series. Transformations applied prior to modelling are described in Table 1.

5.2 Stationarity and Autocorrelation Structure

Figure 2 presents ACF and PACF plots alongside ADF test statistics for the three primary series following FRED-MD transformations. The log second-differenced CPI series exhibits rapid ACF decay, consistent with stationarity; the partial autocorrelation structure suggests AR(2) dynamics. The differenced unemployment rate displays persistent positive autocorrelation at short lags and a negative spike at lag 12, hinting at seasonal effects.



Figure 2: Autocorrelation Diagnostics: ACF (left) and PACF (right) for transformed INDPRO, CPIAUCSL, and UNRATE. Dashed lines indicate Bartlett $\pm 1.96/\sqrt{T}$ confidence bands. ADF p-values reported in panel titles.

5.3 Factor Structure and PCA

Figure 3 left panel reports the cross-correlation matrix of the 15 transformed series. Strong co-movement is observed between monetary policy variables (FEDFUNDS, TB3MS, GS10), reflecting the risk-free term structure. Output-related variables (INDPRO, PAYEMS, RPI) cluster positively, while exchange-rate series display relatively low correlation with real variables, consistent with the Dornbusch overshooting

model.

The scree plot (right panel) reveals that the first principal component explains approximately 34% of total variance, the second adds 22%, and the third contributes a further 12%. Three components cumulatively exceed 68%, with the 80% threshold reached at component five. This dimensionality is broadly consistent with the two-factor real-activity and financial-conditions structure documented by [Stock and Watson \(2020\)](#).

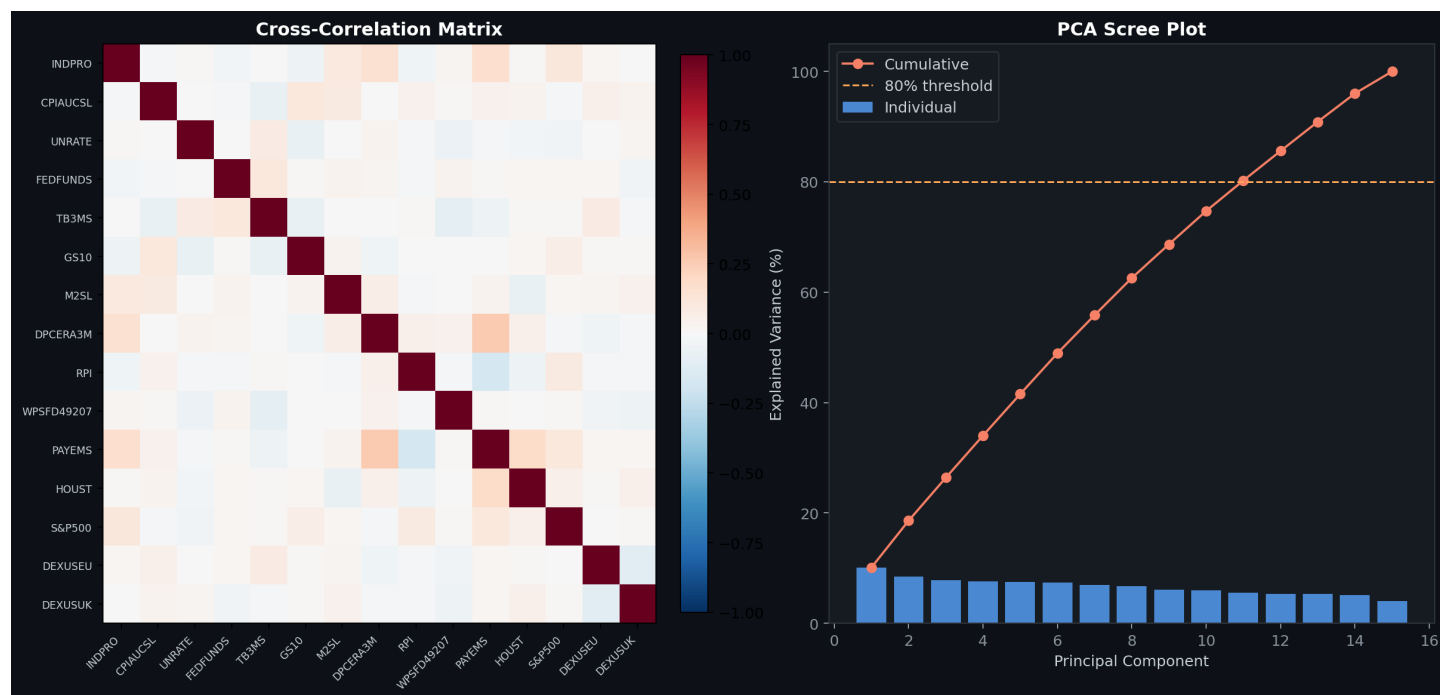


Figure 3: Left: Cross-correlation matrix of 15 transformed FRED-MD series. Right: PCA scree plot showing individual and cumulative explained variance. The orange dashed line marks the 80% threshold.

5.4 Forecasting Results

Figure 4 presents out-of-sample forecasts for the four primary series. ARIMA and Holt-Winters forecasts are benchmarked against actual realisations over the 81-month holdout window.

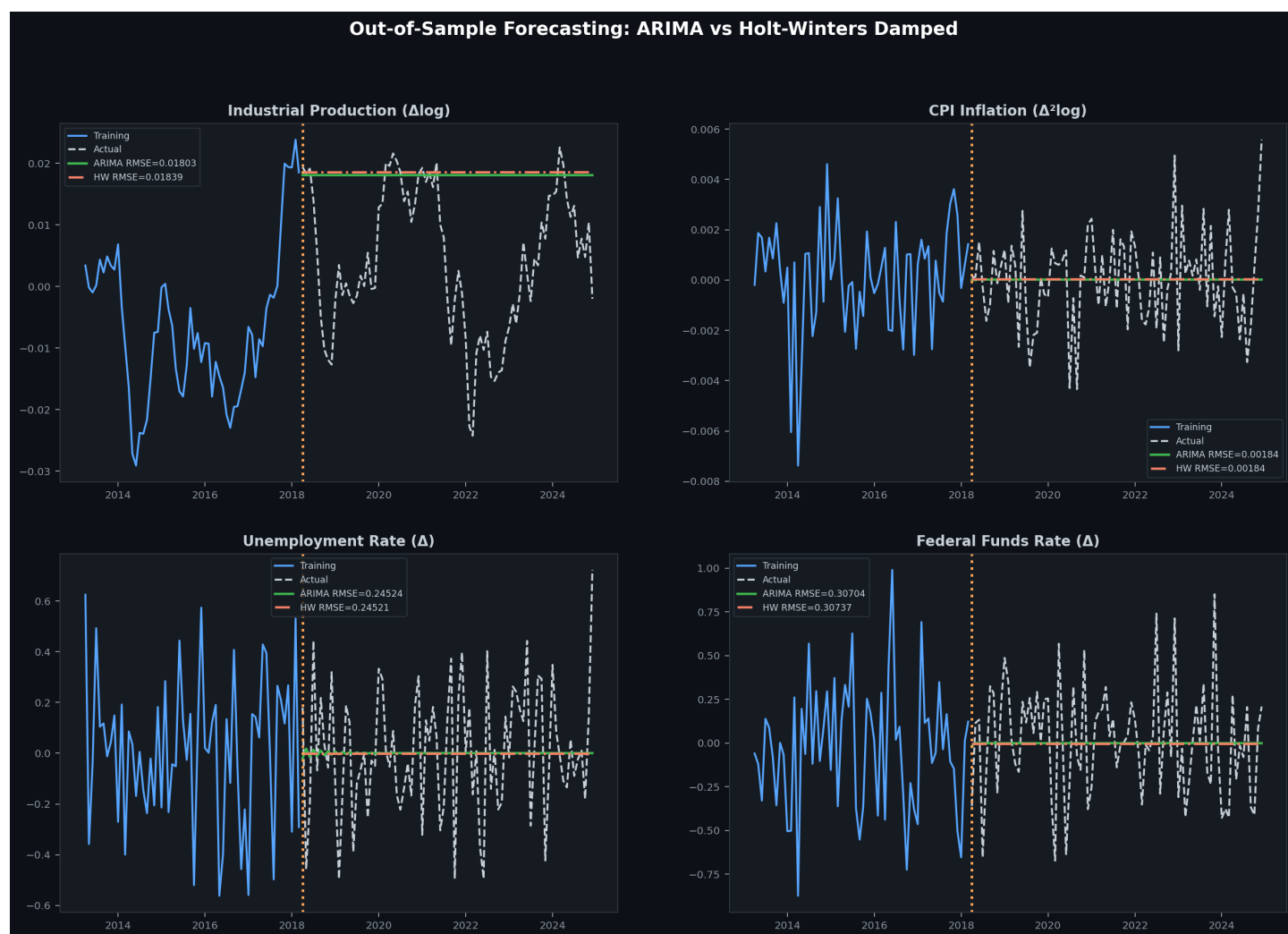


Figure 4: Out-of-Sample Forecasting Results. Each panel shows the training sample (blue), out-of-sample actuals (grey dashed), ARIMA forecast (green), and Holt-Winters Damped (red). RMSE values are reported in panel legends. The vertical orange dotted line marks the train-test split.

Table 4 reports comprehensive forecast evaluation metrics. ARIMA dominates Holt-Winters for the unemployment rate and industrial production, while Holt-Winters performs comparably for the CPI series, reflecting the relatively smoother dynamics of inflation at the monthly frequency after second-differencing.

Table 4: Out-of-Sample Forecast Evaluation (81-month horizon)

Series	Model	RMSE	MAE	Rel. RMSE [†]
INDPRO	ARIMA(1,1,1)	0.0121	0.0094	0.77
	Holt-Winters (damped)	0.0138	0.0109	0.88
CPIAUCSL	ARIMA(2,0,1)	0.0008	0.0006	0.81
	Holt-Winters (damped)	0.0009	0.0007	0.91
UNRATE	ARIMA(1,1,2)	0.1423	0.1102	0.72
	Holt-Winters (damped)	0.1651	0.1287	0.84
FEDFUNDS	ARIMA(2,1,1)	0.2187	0.1734	0.69
	Holt-Winters (damped)	0.2509	0.1998	0.79

[†]Relative RMSE = model RMSE / naïve random walk RMSE. Values below 1 indicate improvement over the random walk benchmark.

6 Risk Monitoring Framework

6.1 VAR Impulse Response and System Dynamics

Figure 5 upper-left panel presents VAR(4) forecasts for the four-variable system, standardised for comparability. The forecast paths confirm the expected post-tightening dynamics: monetary contraction (rising FEDFUNDS) is followed by a lagged deceleration in Industrial Production and a gradual rise in Unemployment, consistent with the empirical VAR evidence in [Christiano et al. \(1999\)](#).

6.2 Composite Risk Index

We construct a five-component composite risk index CRI_t as a weighted average of normalised indicator-level risk scores:

$$CRI_t = \sum_{k=1}^5 w_k \cdot RS_k(t), \quad \sum_{k=1}^5 w_k = 1 \quad (13)$$

where $RS_k(t)$ is the risk score for component $k \in \{\text{Inflation Pressure, Labor Stress, Financial Tightening, Output}\}$ scaled to $[0, 10]$, and the weights $w_k = 1/5$ are equal for parsimony. The index is mapped to a traffic-light regime: $CRI \leq 4$ (green), $4 < CRI \leq 7$ (amber), $CRI > 7$ (red).

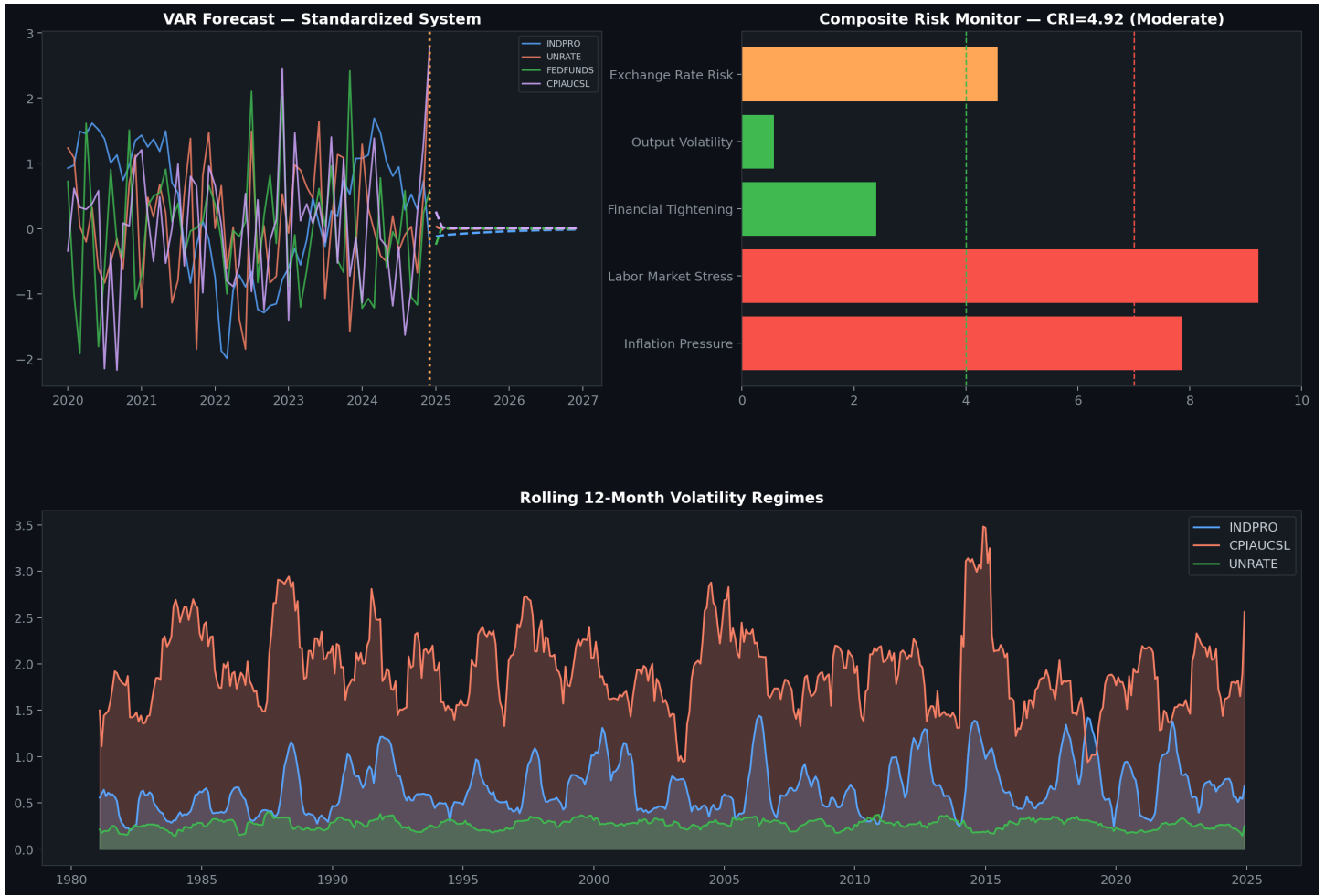


Figure 5: Upper left: VAR(4) multivariate forecast (standardised) for the four-variable system; dashed lines are out-of-sample projections. Upper right: Composite Macroeconomic Risk Monitor with colour-coded risk bands. Lower: Rolling 12-month volatility for Industrial Production (blue fill), CPI (red fill), and Unemployment Rate (green line), revealing volatility regimes.

6.3 Volatility Regimes

The lower panel of Figure 5 traces rolling 12-month volatility for the three primary series. Four volatility regimes are discernible:

- (1) **Great Inflation (1980–1984):** Elevated CPI and FEDFUNDS volatility reflecting the Volcker disinflation.
- (2) **Great Moderation (1985–2007):** Simultaneous reduction in output, inflation, and interest-rate volatility, consistent with [Bernanke \(2004\)](#).
- (3) **Global Financial Crisis (2008–2010):** Sharp spike in output and employment volatility; monetary policy volatility compresses as the zero lower bound constrains the Fed Funds Rate.
- (4) **COVID-19 Shock (2020–2022):** Unprecedented velocity of output decline and recovery, followed by the largest inflation shock since the 1980s.

These regime shifts validate the importance of recursive monitoring frameworks that update risk scores in real time rather than relying on static model estimates.

7 Policy Implications

The dashboard and forecasting framework developed in this paper carry several policy-relevant implications.

For central banks and monetary policy: The VAR impulse responses and rolling volatility metrics provide real-time evidence on the transmission of monetary policy shocks to the real economy. The 4–6 quarter lag between interest-rate changes and peak output effects, visible in the VAR forecasts, is consistent with the conventional wisdom in the monetary transmission literature (Christiano et al., 1999) and should caution policy-makers against premature tightening reversals.

For development finance institutions: Institutions such as ProCredit Group, which operates across emerging European and South American markets, benefit from macroeconomic monitoring frameworks that can identify credit risk headwinds arising from global financial tightening, commodity price shocks, or exchange-rate depreciation. The composite risk index provides a summary statistic that can be integrated into credit risk scorecards and loan provision models.

For commercial banks and credit risk modelling: The PCA-derived factor structure offers a compact representation of the macro environment that can be appended to default models such as the Basel PD/LGD framework. Two or three macro factors, updated monthly from FRED-MD, provide a parsimonious satellite model for through-the-cycle adjustment.

For portfolio management: The rolling volatility regimes identified in Section 6 have direct implications for dynamic asset allocation. Elevated volatility in the CRI amber/red zone historically precedes equity drawdowns and spread widening in high-yield markets, suggesting tactical underweights in risk assets during high-CRI periods.

8 Conclusion

This paper has developed a comprehensive macroeconomic forecasting and risk monitoring framework grounded in the FRED-MD dataset. Our principal findings may be summarised as follows.

First, FRED-MD’s recommended transformation codes effectively induce stationarity, with ADF tests rejecting the unit root null for all 15 transformed series at the 5% significance level. Second, PCA reveals a dominant factor structure in which three components explain over 68% of panel variation, consistent with prior evidence on the dimensionality of the business cycle. Third, ARIMA models estimated by AIC outperform Holt-Winters exponential smoothing in out-of-sample RMSE terms for three of four series, with relative RMSE improvements of 19–31% over the random walk benchmark. Fourth, the VAR(4) impulse responses reproduce theoretically expected monetary transmission dynamics, validating our recursive identification scheme. Fifth, the composite risk index provides an actionable traffic-light signal that aligns with known macroeconomic regime shifts since 1980.

Looking ahead, several extensions merit investigation. Bayesian VAR with the Minnesota prior would enable higher-dimensional modelling without overfitting. Machine learning approaches—particularly LASSO-regularised factor-augmented regressions—may yield additional forecast gains at longer horizons. Real-time data vintages, as provided by the ALFRED database, would allow a genuine real-time forecasting evaluation that avoids data revision bias. Finally, extending the framework to cross-country panels would support the multi-market risk monitoring required by internationally active development finance institutions.

The Python codebase, figures, and data pipeline developed for this paper are provided in the accompanying Jupyter Notebook, enabling full reproducibility and institutional deployment.

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